Step 0: Install Python Version 3

https://realpython.com/installing-python/

Step 1: Install Scikit-Learn

First, we need to install our dependency, the Python library scikit-learn(sklearn) which is a free software machine learning library for the Python Programming Language. The random library is part of the python standard libraries so we don't need to install it, it's already available for us !

pip install scikit-learn.

Step2: Create Your Training dataset

Usually if you have a data set, you would want to split it into a training set, validation set, and testing set. Since this is a simple example we are creating our own training set to train our linear regression model with. The training set will contain the input and expected output values of the data set that I want to train the model on. The input will be random integer values created by the library Random we will use the method randint(a,b) , which returns a random integer N such that a <= N <= b. The output will be determined by a function that we will create.Recall a function usually called 'f' takes in some input/parameter value usually called 'x' and produces some output value usually represented by 'y', such that f(x) =y. The function that I will create will take in multipleparameters so not just 'x' but 'a', 'b', and 'c' instead such that

we get the following function.

f(a,b,c,d) = 10a + 2b + 3c +4d = y

Example of how this function works, if we had

values a=1, b=2, c=20 then y would equal 74 like the following:

f(1,2,20) = 10(1) + 2(2) + 3(20) = 74

Example Of Training Set:

Input			Output
a	b	c	y
1	2	20	74
10	5	2	116
5	4	8	82
4	6	9	79
1	2	3	23
0	1	3	11

Once we have created the training set, we will split each row into two lists (a input training set , and it's corresponding output training set). So now we have two lists of all the inputs and their corresponding outputs.

Code To Generate Training Set:

```
from random import randint
TRAIN_SET_LIMIT = 1000
TRAIN_SET_COUNT = 100
TRAIN_INPUT = list()
TRAIN_OUTPUT = list()
for i in range(TRAIN_SET_COUNT):
  a = randint(0, TRAIN_SET_LIMIT)
  b = randint(0, TRAIN_SET_LIMIT)
  c = randint(0, TRAIN_SET_LIMIT)
  d = randint(0, TRAIN_SET_LIMIT)
  op = (7*a) + (3*b) + (4*c) +(9*d)
  TRAIN_INPUT.append([a, b, c, d])
  TRAIN_OUTPUT.append(op)
```

Step 3: Train Our Linear Regression Model

Next we will use the method LinearRegression() from the Python library scikit-learn to train and create our model. This model will try to “model” the function that we created for the training dataset.

Note: When we ‘fit’ a function that is just another word for training.

```
from sklearn.linear_model import LinearRegression

predictor = LinearRegression(n_jobs=-1)

predictor.fit(X=TRAIN_INPUT, y=TRAIN_OUTPUT)
```

Step 4: Test Our Linear Regression Model Now let’s see if our LinearRegression function can approximate the function that we created and give us an accurate prediction (the correct answer).

```
 X = [[10, 20, 30, 40]]
```

The outcome should be 7*10 + 20*3 + 30*4 +9*40 = 610. Let’s see what we got…

```
X_TEST = [[10, 20, 30, 40]]

outcome = predictor.predict(X=X_TEST)

coefficients = predictor.coef_

print('Outcome : {}\nCoefficients : {}'.format(outcome, coefficients))
```

Outcome : [610.]

Coefficients : [7. 3. 4. 9.]

Did you notice what just happened? The model accessed the training data, which it used to calculate the weights (coefficients) to assign to the inputs to arrive at the correct output. When giving the model test data, it successfully managed to get the correct answer!

```
# Import the libraries from random import randint

 from sklearn.linear_model import LinearRegression

# Create a range limit for random numbers in the training set, and a count of the number of rows in the training set

TRAIN_SET_LIMIT = 1000

TRAIN_SET_COUNT = 100
```

Create an empty list of the input training set 'X' andcreate an empty list of the output for each training set 'Y'

TRAIN_INPUT = list()

TRAIN_OUTPUT= list()

#Create and append a randomly generated data set to the input and output

for i in range(TRAIN_SET_COUNT):

 a = randint(0, TRAIN_SET_LIMIT)

 b = randint(0, TRAIN_SET_LIMIT)

 c = randint(0, TRAIN_SET_LIMIT)

 d = randint(0, TRAIN_SET_LIMIT)

 #Create a linear function for the output dataset 'Y'

op = (7*a) + (3*b) + (4*c) +(9*d)

TRAIN_INPUT.append([a,b,c,d])

TRAIN_OUTPUT.append(op)

predictor = LinearRegression(n_jobs=-1) #Create a linear regression object NOTE n_jobs = the number of jobs to use for computation, -1 means use all processors predictor.fit(X=TRAIN_INPUT, y=TRAIN_OUTPUT) (approximate a target function) #fit the linear model X_TEST = [[10,20,30,40]] #Create our testing data set, the ouput should be 7*10 + 3*20 + 4*30+9*40 = 610

outcome = predictor.predict(X=X_TEST) # Predict the ouput of the test data using the linear model

coefficients = predictor.coef_ #The estimated coefficients for the linear regression problem.

print('Outcome: {} \n Coefficients: {}'.format(outcome, coefficients))